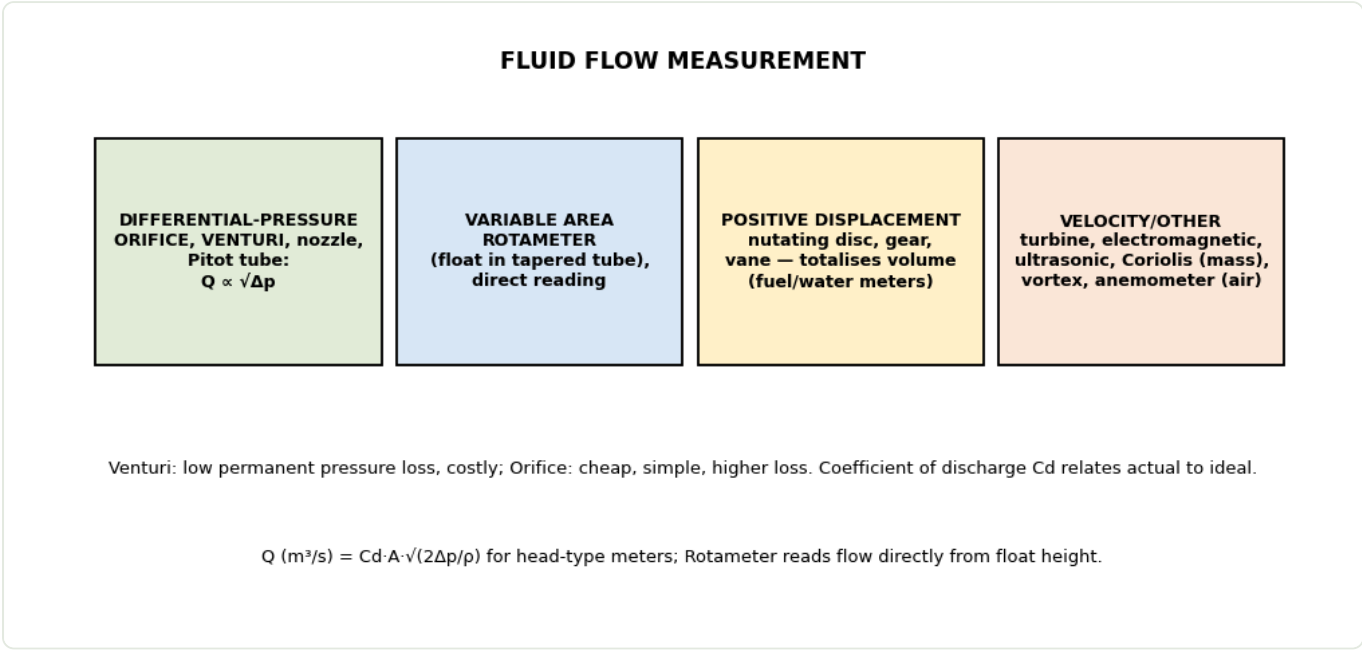
UNIT 9 · TOPIC 7

Measurement of Fluid Flow Rates

Syllabus topic: Measurement of fluid flow rates



Fluid flow measurement — meter families

Differential-pressure (head) meters

$$Q = Cd \cdot A \cdot \sqrt{(2\Delta p/\rho)} \rightarrow \text{flow} \propto \sqrt{\Delta p}$$

A constriction creates a pressure drop; the coefficient of discharge Cd corrects ideal flow to actual flow.

Meter	Features
ORIFICE plate	Cheap, simple; HIGH permanent pressure loss
VENTURI	Gradual convergent–divergent; LOW permanent loss; accurate but costly
Flow NOZZLE	Intermediate
PITOT tube	POINT VELOCITY from stagnation vs static pressure

Other flow meters

Meter	Principle / use
ROTAMETER (variable area)	A float rises in a tapered vertical tube until drag balances weight; the height reads flow directly
POSITIVE-DISPLACEMENT (nutating disc, gear, vane)	Totalise volume — fuel and water meters
Turbine	Rotor speed ∝ velocity
Electromagnetic	Conductive liquids
ULTRASONIC; vortex-shedding	Non-intrusive / velocity-based
CORIOLIS	Direct MASS flow
ANEMOMETER	Air/gas velocity

Practice MCQs

Q63. Head-type flow meters give flow rate proportional to:

(a) Δp (b) $\sqrt{\Delta p}$
(c) Δp^2 (d) $1/\Delta p$

Q64. Which flow meter has the LOWEST permanent pressure loss?

(a) Orifice (b) Venturi
(c) Flow nozzle (d) None

Q65. The cheapest and simplest differential-pressure flow element is the:

(a) Venturi (b) Orifice plate
(c) Coriolis (d) Turbine

Q66. A rotameter is a ___ flow meter:

(a) Differential-pressure (b) Variable-area (float in tapered tube)
(c) Positive-displacement (d) Ultrasonic

Q67. A Pitot tube measures:

(a) Mass flow (b) Point velocity (stagnation vs static pressure)
(c) Temperature (d) Density

Q68. Which meter measures MASS flow rate directly?

(a) Orifice (b) Coriolis
(c) Rotameter (d) Pitot

Q69. Fuel or water consumption is totalised by a:

(a) Positive-displacement meter (b) Pyrometer
(c) Thermocouple (d) Accelerometer

Q70. The coefficient of discharge (Cd) of a flow meter relates:

(a) Actual to ideal flow (b) Pressure to temperature
(c) Mass to volume (d) Sound to vibration

Answers

Q63 — (b) $\sqrt{\Delta p}$ — *Head meters: $Q \propto \sqrt{\Delta p}$.*

Q64 — (b) Venturi — *Venturi — lowest loss.*

Q65 — (b) Orifice plate — *Orifice — cheapest.*

Q66 — (b) Variable-area (float in tapered tube) — *Rotameter = variable area.*

Q67 — (b) Point velocity (stagnation vs static pressure) — *Pitot = velocity.*

Q68 — (b) Coriolis — *Coriolis = mass flow.*

Q69 — (a) Positive-displacement meter

Q70 — (a) Actual to ideal flow

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney; measuring-instruments course notes (as given in the source notes, July 2026); standard fuel-testing methods (ASTM/IS); NPTEL measurement lectures.